Substantive Content Corrections

Replacement text and tables for the keratoconus report; prepared from the supplied PDF and the implemented dashboard artifacts. Layout and image-placeholder corrections are intentionally out of scope.

Scope and publication status

This document corrects claims that are not supported by the current implementation or by a matched multimodal evaluation. It is a **research-prototype** report, not evidence of a clinically validated diagnostic system. The source PDF should not claim a trained, tested, or deployed multimodal staging/progression/RAG system unless each component is implemented and evaluated on patient-eye matched data.

Most important correction: the image model and clinical XGBoost model were trained on different cohorts and are not linked at patient-eye level. Their probabilities may be displayed together in the dashboard, but that is **late probability averaging**, not a validated multimodal model. No fused accuracy, AUC, sensitivity, specificity, staging, progression, or asymmetry result should be reported.

1. Author-facing correction table

Original claim	Correct statement	Required action
Early/subclinical KC staging, progression risk, and inter-ocular asymmetry are predicted.	The deployed system is binary: keratoconus versus non-keratoconus. It has no trained or validated stage, progression, or asymmetry head.	Remove multi-head claims, Head 2/3 equations, and all stage/progression/asymmetry dashboard outputs. List them as future work only.
A learned multimodal fusion combines images, spatial grids, and biomechanics.	The dashboard accepts seven image maps and 12 structured clinical inputs. It combines available branch probabilities using a user-selected image weight. Raw numerical map grids are not ingested.	Rename as dual-stream decision support with unvalidated late fusion. Do not call it a trained multimodal model until matched-data OOF stacking is evaluated.
RAG, FAISS/LangChain, Integrated Gradients, and Attention Rollout are part of the deployed system.	The current FastAPI dashboard provides Grad-CAM for supplied images and TreeSHAP-style XGBoost feature contributions. It does not implement retrieval, FAISS, LangChain, Integrated Gradients, or Attention Rollout.	Remove these completed-system claims; retain only as future design goals if desired.
Three independent datasets establish the proposed fusion.	The three datasets support separate, heterogeneous baseline experiments. They do not establish a common, matched multimodal cohort or fused-model performance.	Report them in separate subsections; delete comparative proposed-system performance claims.
External validation establishes deployability.	The mapped external test produced AUROC 0.9508 but the imported threshold yielded 100% sensitivity and 0% specificity. It is exploratory domain-shift analysis only and not usable clinical validation.	Do not report its accuracy as evidence of external deployment performance.

2. Replacement abstract

Abstract. Keratoconus is a progressive corneal ectasia for which automated analysis of corneal maps and structured measurements may support research screening workflows. We developed a research dashboard that accepts up to seven corneal map types and a 12-variable clinical/topographic feature set. The image branch is an EfficientNet-B0 encoder with seven-map attention fusion; the clinical branch is an out-of-fold calibrated XGBoost model. The dashboard returns branch-specific probabilities, a manually controlled late-fusion probability when both are available, Grad-CAM overlays for provided image maps, and tree-based clinical feature contributions. The image and clinical branches were trained on different cohorts; therefore, the present work does not report a validated multimodal performance estimate.

On the independent image test cohort (150 cases, binary keratoconus versus non-keratoconus), the EfficientNet-B0 model with test-time augmentation achieved 94.0% accuracy, 0.9904 AUROC, and 0.9290 macro-F1; keratoconus sensitivity was 82.0% and non-keratoconus specificity was 100.0%. The clinical-model results require a deduplicated patient-eye rerun before they are used as publication-grade evidence. This work is a research prototype and not a diagnostic device.

3. Corrected system diagram and caption

Inputs	Branch model	Current output
Up to seven image maps: CT_A, EC_A, EC_P, Elv_A, Elv_P, Sag_A, Sag_P	EfficientNet-B0 shared encoder + seven-map attention fusion	Image probability; Grad-CAM for supplied maps
12 clinical/topographic values	Calibrated XGBoost	Clinical probability; tree-based feature contributions
Both branch probabilities, if supplied	User-selected late probability average	Combined research score and disagreement warning

Figure 1 replacement caption. Current dashboard architecture. Image and clinical branches are evaluated independently. When both are supplied, the dashboard reports a configurable probability average; this averaging is not a learned or clinically validated multimodal fusion model. Missing image maps are zero-filled at inference and should be treated as a data-quality limitation, not evidence of validated missing-modality robustness.

4. Replacement methodology

Study design. This is a retrospective, cross-sectional research prototype. All datasets must be described separately because they have different label definitions, scanners, data formats, and patient linkage. No claim of multimodal fusion performance is made unless image and clinical features are paired to the same patient eye before partitioning.

Image branch. The implemented image model is a binary EfficientNet-B0 architecture. Each available map is resized and ImageNet-normalized, encoded by a shared EfficientNet-B0 backbone, projected to a 320-dimensional embedding, and combined across the seven map tokens by multi-head attention. The classifier predicts keratoconus versus non-keratoconus. The training notebook maps Normal and Suspect cases to the non-keratoconus class. Augmentation includes resizing, horizontal flips, rotations, and affine translation. Test-time augmentation averages logits across four deterministic transforms. Therefore, the manuscript must not present this branch as a three-class Normal/Suspect/Keratoconus classifier.

Clinical branch. The clinical model uses age, gender, astigmatism magnitude and axis, Kmax and axis, central and thinnest pachymetry, thinnest-point coordinates, and anterior/posterior asphericity. The stored XGBoost model uses imputation and an out-of-fold fitted calibration model. A future publication-grade rerun must first resolve repeated clinical records to one record per patient eye, correct identifier aliases, encode axis variables circularly, maintain patient-grouped partitions, and select thresholds only from development out-of-fold predictions.

Fusion protocol required before claiming multimodality. Create one matched table per patient eye; split patients before preprocessing; generate image and clinical out-of-fold logits from the same folds; train the fusion model only on those out-of-fold logits; calibrate and choose its operating threshold using development out-of-fold data; then evaluate once on an untouched patient-held-out cohort. A regularized logistic stacker should replace manual fixed weighting. Report branch-only and fused results with patient-clustered confidence intervals.

Explainability. Report Grad-CAM as a localization aid for the image model and tree-based feature contributions for the clinical model. Neither should be interpreted as causal evidence. Do not claim Integrated Gradients or Attention Rollout unless those methods are implemented, checked for completeness/sanity, and included in the dashboard.

5. Corrected results tables

The following table replaces Tables 4, 5, 6, 7, and 8 insofar as they make claims about the currently implemented system. It separates results that are directly supported by artifacts from results that must be rerun before publication.

Experiment	Cohort / label	Supported result	Publication interpretation
Image branch, independent test with TTA	150 cases; binary KC vs non-KC; 50 KC, 100 non-KC	Accuracy 0.9400; AUROC 0.9904; macro-F1 0.9290; KC recall 0.8200; non-KC recall 1.0000	Report as a binary image-model result only. It does not establish early/subclinical detection or multiclass staging.
Clinical XGBoost artifact	Clinical/topographic records with repeated examinations retained	Stored report: locked-test accuracy 0.9828; AUROC 0.9993; sensitivity 0.9646; specificity 0.9944	Do not use as final publication evidence until the one-eye-per-record deduplication and patient-level evaluation are rerun. Repeated exams otherwise over-weight some eyes.
Mapped external image test	300 cases; exploratory scanner/domain mapping	AUROC 0.9508; at imported threshold: accuracy 0.5000, sensitivity 1.0000, specificity 0.0000	Report only as a threshold-transfer failure / domain-shift warning; not as external clinical validation.
Fused image + clinical model	No matched patient-eye cohort evaluated	Not available	Do not report a fused accuracy, AUC, staging, or advantage over baselines.

6. Replacement dashboard description

The dashboard is a FastAPI service with a built web frontend, rather than a Streamlit-only deployment. Users may submit image maps, clinical values, or both. The dashboard validates required clinical fields, reports a warning when fewer than seven image maps are supplied, returns branch-specific scores and a combined research score, and flags branch disagreement. It displays Grad-CAM overlays only for uploaded image maps and clinical feature contributions only when clinical input is supplied. The interface must state that the combined score is not a validated multimodal diagnostic output and that a clinician must independently review the source examinations.

7. Replacement conclusion and future work

This work delivers a dual-stream research dashboard for keratoconus decision support: a seven-map EfficientNet-B0 image classifier and a calibrated structured clinical XGBoost classifier. The image model showed strong binary discrimination in its independent test cohort, while the clinical model indicates that structured topographic measurements contain substantial signal. The current evidence does not support claims that the system detects early or subclinical disease, performs multiclass staging, estimates progression risk, detects inter-ocular asymmetry, implements RAG-grounded clinical reasoning, or achieves validated multimodal performance. These components remain future work.

Priority next steps are: (1) create a de-duplicated, patient-eye matched image-clinical cohort; (2) pre-register a patient-level split and locked evaluation protocol; (3) train and calibrate an out-of-fold late-fusion model; (4) validate prospectively across sites and scanner vendors; (5) separately collect longitudinal follow-up data before modelling progression; and (6) perform clinician review of all explanation and interface outputs. Any regulatory, treatment, or cross-linking recommendation must remain outside the model output unless backed by an appropriate clinical study and governance process.

8. References and citation corrections

Retain only verifiable, directly relevant sources in the final manuscript. The related-work section should focus on keratoconus imaging, structured topography/tomography, biomechanics, calibration, explainability, and clinical AI evaluation. Remove unrelated brain-tumour, stroke, prescription-recognition, and generic IoT-monitoring citations; they do not establish a keratoconus research gap. Verify every remaining DOI, title, author list, venue, and date against the publisher or PubMed before submission.

Internal evidence used for these corrections: supplied report PDF; *model1.ipynb* (binary image-model training and independent-test output); *journal_xgb_results/metrics.json* (clinical artifact metrics and repeated-examination note); *external_validation_results/external_validation_metrics.json* (exploratory threshold-transfer result); and the FastAPI dashboard source files (*server.py*, *utils/model_loader.py*, and *utils/inference.py*).